

Write a Python function to analyze the histogram of the given input image based on color levels using Open CV.

Code:

```
import cv2, urllib.request

import matplotlib.pyplot as plt

# Download sample image

urllib.request.urlretrieve(

    "https://raw.githubusercontent.com/opencv/opencv/master/samples/data/lena.jpg",

    "img.jpg"

)

# Read image

img = cv2.imread("img.jpg")

# Plot histogram for B, G, R channels

for i, c in enumerate(['b','g','r']):

    hist = cv2.calcHist([img],[i],None,[256],[0,256])

    plt.plot(hist, color=c)

# Display histogram

plt.title("Color Histogram")

plt.xlabel("Pixel Value")

plt.ylabel("Frequency")

plt.show()
```

Output:

